

Week 13 Recitation

TA Suha Kim

ECON2001.01 (Prin. Micro.)

Apr 4, 2025

Monopoly

Monopoly

- A market with a single seller
- There is no close substitute for the good.
- The monopolist (seller) has a **Market power**: Can **set the price (Price-maker)**.

Source of Market Power - Entry barrier

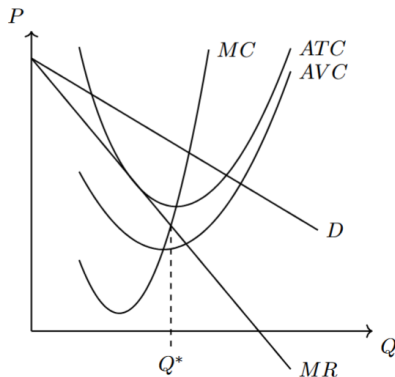
- Legal market barriers
E.g. Patents, Copyright
- Natural market power
 - The monopolist owns or controls a key resource necessary for production
 - Economies of Scale: Decreasing ATC - e.g. Utility companies

Monopoly

Monopolist's Problem

Monopolist's Problem - Maximizing Profit

- Chooses a quantity Q^* at $MR = MC$.
 - Unlike the perfectly competitive market, $P \neq MR$.
 MR decreases in Q .



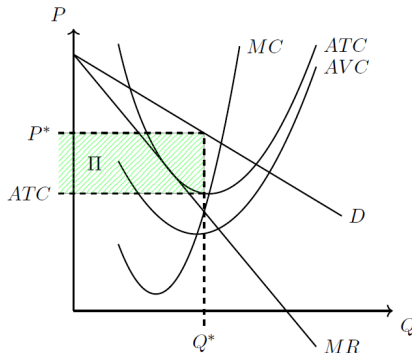
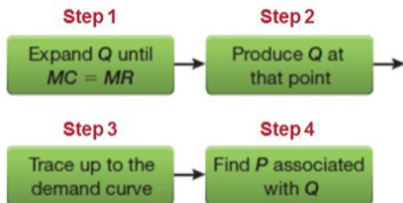
Monopoly

Monopolist's Problem

Monopolist's Problem - Maximizing Profit

- Chooses a quantity Q^* at $MR = MC$.
 - Unlike the perfectly competitive market, $P \neq MR$.
 MR decreases in Q .
 - Because a monopolist is the only seller in the market, it faces the market demand curve (downward sloping)
 - Therefore, the monopolist's profit function can be written as $\Pi = P \cdot Q(P) - C(Q)$
 - Revenue is derived from $Revenue = P(Q) \cdot Q$
(The quantity demanded in the market depends on the price in the market)
 - Marginal revenue will then be equal to, $MR = \frac{\Delta P}{\Delta Q} \cdot Q + P(Q)$
 - By definition, the MR curve is **twice as steep** as the demand curve
- Set price equal to consumer's WTP of the last unit produced.
 $\Rightarrow P^* =$ the height of the Demand curve at Q^*

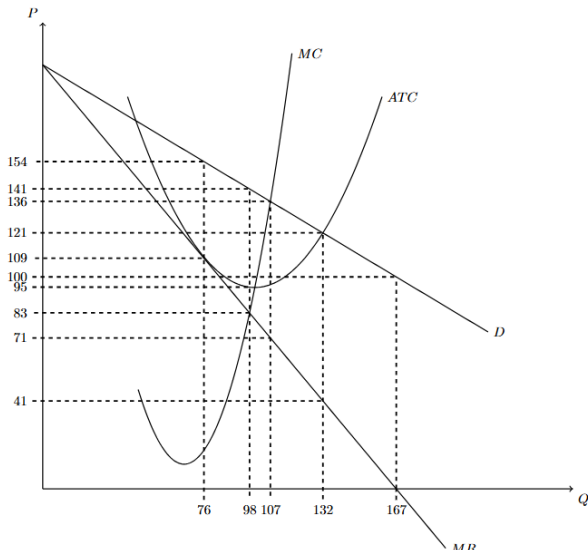
Monopolist's Problem - From the graph



Monopoly

Monopolist's Problem - Exercise (Problem Set Chap 12, #3.)

Figure 2: Market for Mjöltnirs



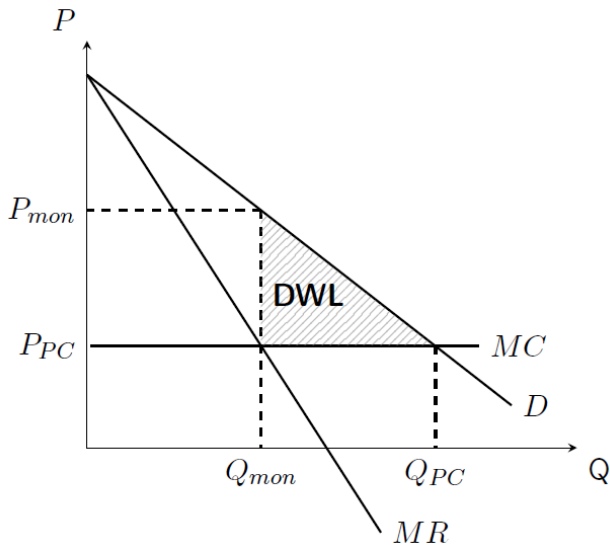
Inefficiency of a monopoly

Perfectly competitive market vs. Monopoly

- Perfectly competitive markets without market distortions
 - Social surplus is maximized $\Rightarrow$ Pareto Efficient
 - $P = D = MC$. ($\leftarrow$ Notice!: Demand (D) represents MB)
- Monopoly
 - $P = D > MC \Rightarrow$ Not Pareto Efficient. (= Broken Invisible hand)
 - Social surplus is not maximized (DWL exists).
 - Under-production than socially optimal level.

Inefficiency of a monopoly

Deadweight Loss

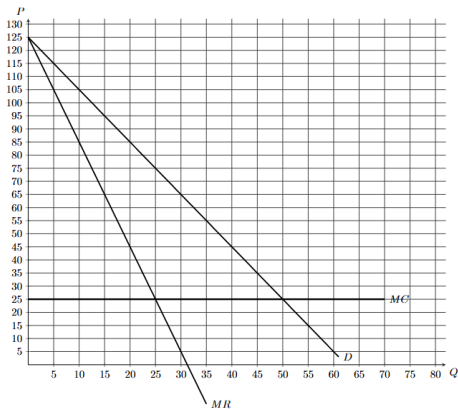


Inefficiency of a monopoly

Exercise (Problem Set Chap 12, #4.)

4. The figure below displays the demand, marginal revenue, and marginal cost curves of a natural monopoly.

Figure 3: A Natural Monopoly



Inefficiency of a monopoly

Resolving inefficiency

① Incentivize the monopolist: **Price discrimination**

- Why? - Increase profits and producer surplus
 - ① First-degree price discrimination (= perfect price discrimination)
 - ② Second-degree price discrimination
 - ③ Third-degree price discrimination

② Governmental policy

: Laws in place to limit noncompetitive behavior – lower market prices through increased competition

- **Antitrust policies**: regulate and prevent anti-competitive pricing
- **Price regulation**
 - ① At Socially optimal price: $P = MC = D$
 - ② At Fair-return price: $P = ATC = D$

Price Effect vs. Quantity Effect

Assume this monopolist is selling 200 units at a price of \$5 each and wants to increase the quantity it sells to 400 units. It lowers the price to \$4 to increase sales.

